

Aadya Goel

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EDUCATION

Carnegie Mellon University

Aug 2025 – May 2027

MS Artificial Intelligence Engineering – Biomedical Engineering (Research Track) | GPA: 4.00, Dean's Honors

- Biomedical Engineering Department Head's Fellowship recipient; Emerging Leaders Mentor (2025–present) ; BME Forum 2025 Poster Presenter

Vellore Institute of Technology

Jul 2021 – Jul 2025

B.Tech Computer Science, Specialization in Bioinformatics | GPA: 3.8 | Class Rank: 15/174 (Top 8%)

WORK EXPERIENCE

University of Pittsburgh Medical Center (UPMC)

Sep 2025 – Present

Graduate Student Researcher

Pittsburgh, PA

- Building AI/ML and systems biology pipelines integrating transcriptomic, proteomic, and clinical datasets from human and mouse models to study virus-induced endothelial dysfunction in collaboration with NIH, Yale and UPitt researchers.
- Applying single-cell analysis (Cell Ranger, Scanpy, Seurat WNN, CellChat) to large-scale NIH ACTIV4c and Pitt Long COVID cohorts for biomarker discovery; constructing multiscale mathematical models of post-COVID chronic comorbidities.

Prognostic Optimization Group

May 2025 – Present

AI Engineering Intern

Dallas, TX (Hybrid)

- Developing agentic AI pipelines using LLMs, LangChain, and RAG to support clinical decision support and patient triage in a telemedicine platform; integrating FastAPI and Pinecone vector databases for intelligent autonomous patient interactions.

SETV Global – Healthcare Technology Startup

Jan 2025 – Jul 2025

Head of Genomics & Lead, Elderly and Women Care Division

Hyderabad, India

- Led 40-person team to develop Gene-omere, an AI genomic analysis platform for genetic abnormality detection, organism classification, and pathogen matching; collaborated & deployed with AIIMS Delhi and Telangana Government on personalized healthcare applications.
- Analyzed Human Metapneumovirus (HMPV) using Gene-omere achieving >65% efficiency and accuracy; findings submitted to WHO, NIH, and CEPI for vaccine development consideration. Presented at Startup Mahakumbh 2025 to 50,000+ global business leaders & 2024 Global AI Summit in Hyderabad.

Cambridge Centre for International Research (CCIR) – University of Oxford

May 2024 – Jun 2025

Research Scholar

Remote

- Authored and published 'Epilepsy Seizure Detection via EEG Signals using Machine Learning' (ICIITCEE 2025); developed deep learning models for detecting epilepsy in patients aged 3–22 using EEG brain signals and signal processing with 97% precision.
- Selected as 1 of 5 students admitted to the CCIR Future Scholars Program (STEM scholarship); nominated as CCIR Spotlight Scholar for exceptional research and part of their YouTube podcast series to share findings.

Ernst & Young (EY) New Delhi

Aug 2023 – Oct 2023

Tech Consulting Intern, Data Analytics Division

New Delhi, India

- Applied ML algorithms to personalize a healthcare navigation app; contributed as frontend developer (React.js, Flutter) on a CDAC-EY smart city initiative.

RESEARCH & PUBLICATIONS

Drug2Cell: Agentic AI System for In-Silico Drug Response Prediction

Mar 2026 – Present

Carnegie Mellon University

Pittsburgh, PA

- Training a CVAE-GAN on 800K scRNA-seq profiles to predict genome-wide drug perturbation responses for unseen compounds using GNN-based molecular encoding.
- Building end-to-end genomics pipeline (scRNA-seq QC, Harmony batch correction, HVG selection, pathway-grouped attention over 3,000 genes); deploying as LangGraph agentic system with RAG over 20K+ PubMed abstracts.

Identification of Suicidal Behavior using Deep Learning – College Capstone

Jul 2023 – Apr 2025

- Built NLP models to detect suicidal risk patterns in social media & brain fMRI data; deployed chatbot successfully classified adolescents at risk of self-harm.

NUS Singapore – Global Academic Internship Program

Dec 2022 – Jun 2023

Research Intern

Singapore

- Built a deep learning model for Emotion Analysis in Sports achieving 98.22% accuracy; researched stock market manipulation detection using deep learning (pre-print available at ResearchGate).

Selected Publications

- Goel A. et al. 'Epilepsy Seizure Detection via EEG Signals using Machine Learning.' ICIITCEE 2025.
- Goel A. et al. 'Leveraging Artificial Neural Networks for Breast Cancer Detection and Prognosis.' ICCIT 2025.
- Goel A. et al. 'Enhancing Image Feature Matching: ORB & HDBSCAN Integration.' ICCCNT 2024, IIT Mandi.
- AGPI: An AI-Powered Genomic Pathogen Intelligence Platform for Integrated Classification and Therapeutic Targeting. bioRxiv preprint.

SKILLS

Machine Learning, DL & AI: Python, PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, LangChain, RAG, LLMs, FastAPI, OCR, JAX, Spark

Bioinformatics & Genomics: Cell Ranger, Scanpy, Seurat WNN, CellChat, LINCX, NCBI BLAST, Biopython, R, Multi-omic data, Single-cell RNA-seq, Mega, PyMol, PDB, UniProt, ChEMBL, SDF/MOL, FASTA

Data Science: NumPy, Pandas, Matplotlib, OpenCV, SQL, Postgres, Pinecone, Microsoft Azure, AWS, GCP, Git, Docker

Domains: Clinical AI, Computational Biology, Genomics, Bioinformatics, Biomedical Signal Processing, NLP, Computer Vision